

control, smart metering, utilities, and fleet management. The integrated GNSS solution consumes less than 15mW in continuous tracking mode and has RF sensitivity that shortens the time to acquire a position fix. It also provides location data without disrupting the cellular or satellite connections, minimising power usage by reducing the active time of the device.

u-blox
<https://www.u-blox.com>

ADAS innovation with power-efficient automotive SoCs

Renesas Electronics Corporation has expanded its R-Car family of system-on-chips (SoCs) for entry-level advanced driver assistance systems (ADAS). The new series enhance AI



processing capabilities and CPU performance, offering a balance of performance

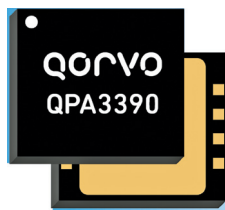
and power consumption for cost-sensitive ADAS applications, such as front smart camera systems, surround-view systems, automatic parking, and driver monitoring systems. Manufactured using 7nm process technology, the series can deliver deep learning performance of up to 17 TOPS, supporting image processing and object recognition through cameras, radar, and lidar. These devices are power-efficient, providing 9 TOPS/W performance, and consume around 5 watts in typical configurations—50 percent less power than comparable products.

Renesas Electronics Corporation
<https://www.renesas.com>

Industry's first 24V power doubler

Qorvo has introduced the industry's first 24V power doubler amplifier for DOCSIS 4.0 broadband and cable TV. This 1.8GHz surface mount module features efficiency and performance, with a footprint that is 30-40% smaller

than hybrid solutions, making it suitable for space-limited applications. It performs well in return loss, operates with low noise, and includes adjustable DC current for control of RF output relative to DC power consumption, making it



suitable for broadband networks. As the only 24V solution on the market, the amplifier delivers 73dBmV total composite power at 11.5W, making it a suitable choice for applications that require low power consumption.

Qorvo
<https://www.qorvo.com>

Highest-powered Wi-Fi HaLow module

Teledatics has introduced and certified its TD-HALOM Wi-Fi HaLow module, known for its low power consumption and long-range capabilities. Powered by Newracom's NRC7394 system-on-a-chip (SoC), this module operates at



1W or 30dBm output power, maximising the range and signal penetration

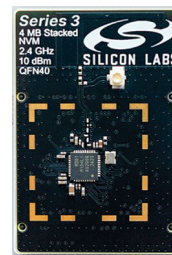
within the limits set by the FCC. The module is designed for applications that require long-range connectivity, such as drone operations, industrial settings, and outdoor environments, where throughput and reliable communication over long distances are essential. Its power output ensures extended connectivity and improved data transfer rates. It is compact, measuring 13.5mm x 18mm and 2.4mm in height, enabling integration into small-scale designs.

Teledatics & Newracom
<https://newracom.com>
<https://www.teledatics.io>

The world's most flexible IoT modem

Silicon Labs has launched the Series 3 SoCs, which feature modems and

memory options that provide processing capabilities for a range of IoT applications, including smart cities,



Industry 4.0, and connected health. Additionally, select Series 3 devices will incorporate Silicon Labs' second-generation matrix vector processor, which improves ML performance in

wireless devices while reducing power consumption. The ongoing data flow between edge devices and the cloud is crucial to IoT operations, improving the roles of edge devices in AI ecosystems by enabling them to make decisions and collect data.

Silicon Labs
<https://www.silabs.com>

Lowest-power AI acceleration co-processor

BrainChip Holdings Ltd has introduced the Akida Pico, a co-processor designed for low-power devices in consumer, healthcare, IoT, defence, and other applications. The Akida Pico



enhances neural network models, leading to an energy-efficient digital design. It supports functions like voice

wake detection, speech noise reduction, and presence detection among others. Operating on less than a milliwatt, it is ideal for battery-powered devices, optimising power usage by activating system processors only when necessary. This makes it effective for continuous monitoring in sensor hubs or systems that require intermittent processing. The Akida Pico is built on the Akida2 platform, suitable for early detection and low-latency tasks in robotics, drones, automotive, and related applications without extensive computing needs.

BrainChip Holdings Ltd
<https://brainchip.com>